Building AI Agents - What we learned -

From LLMs -> Agents -> Multi-Agents -> Real-world SE

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Audience Q&A

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Today’s goals

Review what we covered in Building AI Agents: practice and research examples

Revisit your expectations from the class

What is NEXT?

agent
tools
reasoning
code
software
benchmark
generation
llm
refactoring
engineering
multi
memory
safety
evaluation
acting
planning
api
collaboration
performance

Big questions we covered

1. What is an agent in software engineering?
2. What architectural ingredients make agents effective?
3. When does multi-agent collaboration help?
4. How do agents perform on real SE tasks like refactoring, repair, and API use?
5. What will it take to make agents robust, safe, and sustainable in practice?

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What was your key takeaway from the research papers we discussed?

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1. Foundations & Vision of agentic SE

What are agents and why they matter for SE

- A Survey on Large Language Model Based Autonomous Agents
- Large Language Model-Based Agents for Software Engineering: A Survey
- Agentic Software Engineering: Foundational Pillars and a Research Roadmap

KEY Ideas: what agents are, what roles they play, and what software engineering becomes when agents are first-class actors

2. Agent architecture patterns: reasoning, tool use, and interfaces

How agents work

- ReAct: Synergizing Reasoning and Acting in Language Models
- Toolformer: Language Models Can Teach Themselves to Use Tools
- Gorilla: Large Language Model Connected with Massive APIs
- PAL: Program-aided Language Models
- SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering
- From Tool Orchestration to Code Execution: A Study of MCP Design Choices
- ToolDreamer: Instilling LLM Reasoning Into Tool Retrievers

KEY Ideas: agent performance is not just about model intelligence; it depends heavily on external tools, execution environments, and how the interaction loop is structured.

3. Multi-agent collaboration and division of labor

When is one agent not enough?

- *AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation*
- *MetaGPT: Meta Programming for a Multi-Agent Collaborative Framework*
- *Improving Factuality and Reasoning in Language Models through Multiagent Debate*
- *RefAgent: A Multi-agent LLM-based Framework for Automatic Software Refactoring*
- *Multi-Agent Coordinated Rename Refactoring*
- *MANTRA: Enhancing Automated Method-Level Refactoring with Contextual RAG and Multi-Agent LLM Collaboration*
- *Intelligent Site Reliability Engineering: A Multi-agent LLM Framework for Automated Incident Analysis and Root Cause Determination*

KEY Ideas: multi-agent systems matter when the task naturally decomposes into roles, perspectives, or phases—not just because “more agents” is fashionable.

4. Refactoring and software maintenance as a proving ground for agents

Why software refactoring is a perfect testbed

- *Multi-Agent Coordinated Rename Refactoring*
- *REFACTORBENCH: Evaluating Stateful Reasoning in Language Agents through Code*
- *Together We Are Better: LLM, IDE and Semantic Embedding to Assist Move Method Refactoring*
- *MANTRA: Enhancing Automated Method-Level Refactoring with Contextual RAG and Multi-Agent LLM Collaboration*
- *RefAgent: A Multi-agent LLM-based Framework for Automatic Software Refactoring*
- *A Causal Perspective on Measuring, Explaining and Mitigating Smells in LLM-Generated Code*

KEY Ideas: refactoring is not a toy use case; it exposes the central challenges of agentic SE—state, context, correctness, tool grounding, and collaboration with developers

5. Evaluation, benchmarks, and reality checks

What breaks in practice, and how do we measure it?

- *REFACTORBENCH: Evaluating Stateful Reasoning in Language Agents through Code*
- *Benchmarking Web API Integration Code Generation*
- *SafePro: Evaluating the Safety of Professional-Level AI Agents*
- *Demystifying LLM-Based Software Engineering Agents*
- *Decoding the Configuration of AI Coding Agents: Insights from Claude Code Projects*
- *More with Less: An Empirical Study of Turn-Control Strategies for Efficient Coding Agents*

KEY Ideas: the right question is not "Can the agent solve the task?, but "At what cost, with what reliability, under what controls, and compared to what baseline?"

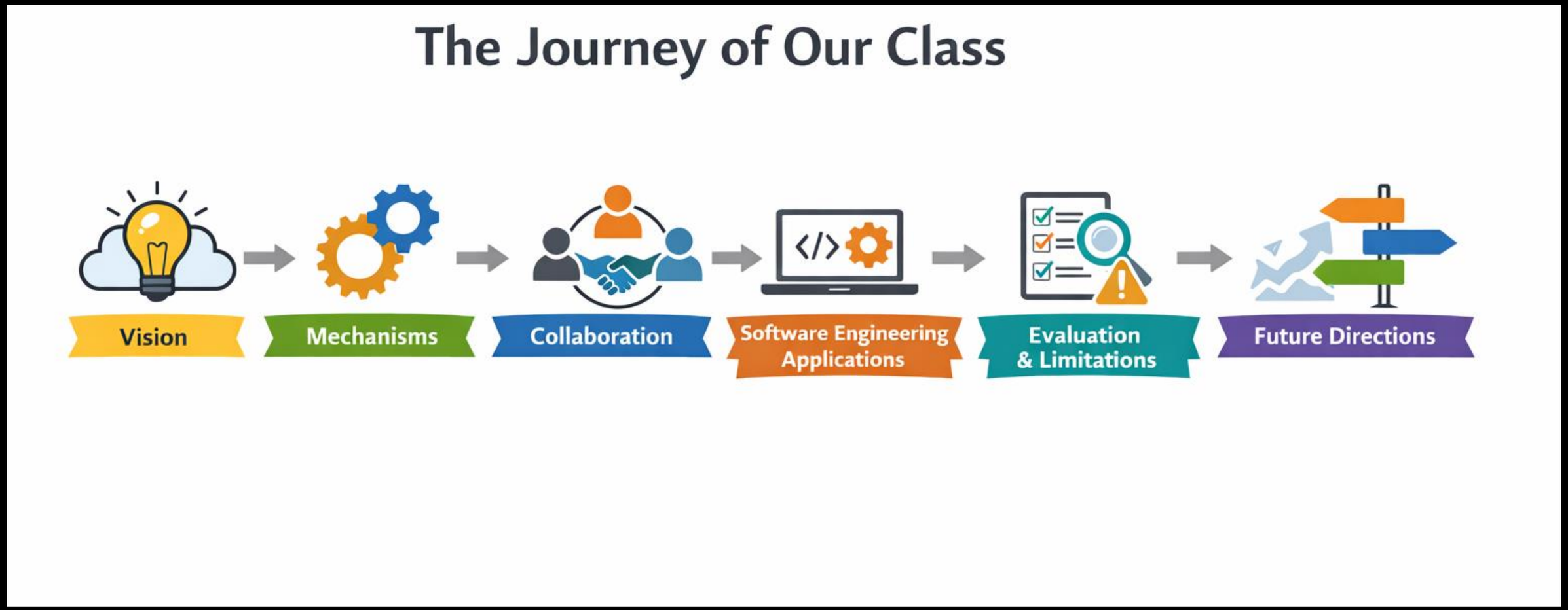
6. Memory, adaptation, skills, and long-horizon robustness

Where the field is going

- *Reflexion: Language Agents with Verbal Reinforcement Learning*
- *MemSkill: Learning and Evolving Memory Skills for Self-Evolving Agents*
- *SoK: Agentic Skills — Beyond Tool Use in LLM Agents*
- *MetaClaw Just Talk – An Agent That Meta-Learns and Evolves in the Wild*
- *AppAgent: Multimodal Agents as Smartphone Users*
- *OpenHands: An Open Platform for AI Software Developers as Generalist Agents*

KEY Ideas: to move from one-shot assistants to durable autonomous systems, agents need mechanisms for memory, reflection, skill reuse, and continual adaptation.

Class Storyline



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Rank the topics you found most interesting/useful/valuable.

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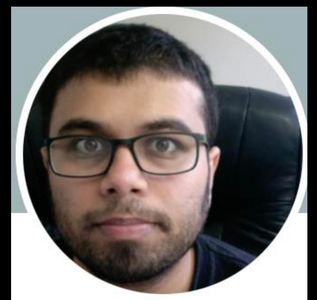
Invited Speakers: Thought leaders from Industry



Ameya Ketkar

Building AI Agents in the industry

Uber gitar

Your growth through research papers

You read 33 cutting-edge research papers
~ equivalent to 700 pages textbook reading

Paper Critiques: **Equipped you with critical thinking**
- learned to evaluate research claims, empirical methods, and limitations in modern Agentic AI research

Research paper presentation: you prepared, delivered, and led class discussion
Equipped you to communicate and connect with the audience

Your growth through hands-on projects

Exercised the whole lifecycle of open-ended, risky project: pitching ideas, forming teams, design, implement, evaluate, and present a novel GenAI SE system

Four awesome research projects:
P1: Agentic Architecture Inference
P2: Cross Repository Pull-Request Review
P3: ReactRefactor Agent
P4: Agentic Tool Discovery in IDE via GUI-Grounded Interaction

Equipped you to lead novel R&D

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How confident do you feel about building Agents in other projects after this course?

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What are your expectations from CSCI 7000-005: Building Agents?

- ☑ A. Do a cool Agentic Project that helps programmers/society
- ☑ B. Learn about exciting Agentic Applications for SE
- ☑ C. Exposure to research on Agentic AI for SE
- ☑ D. Learn about the Challenges in building Agentic software
- ☑ E. Learn how Agentic software differs from classical software
- ☑ F. How to leverage the strengths of each tool
- ☑ G. How academia changes to adapt to AI
- ☑ H. Best practices and ethical use of these tools
- ☑ I. How to make scalable Agentic systems

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What is one expectation you had from this course that you feel was met?

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What you now understand（That most people don’t)

Agents are systems, not models

Tool use is fundamental

Multi-agent is about structure, not scale

SE tasks expose real limitations

Simplicity often wins

Memory & skills are the future

You grew so MUCH

Research papers & critiques

Presentations

Hands-on project work

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How satisfied are you with your learning experience in this course?

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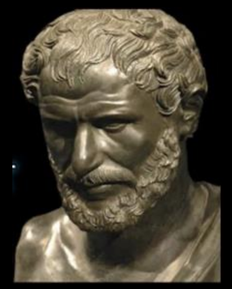
From Users->Builders->Researchers

Change is the only guaranteed constant



"AI **will not** displace programmers. But programmers who use AI **will** displace programmers who do not use AI." – Prof Danny Dig

Agentic AI literacy, proficiency, fluency:
Understand: What are Agents? Why Agents?
Apply: When to use Agents?
Create: How to build Agents?

Together We Go Further

FCQ Evaluation: Help me help students

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What changed in how you approach problems, think, or work with Agentic AI compared to the beginning of the semester? That shift is an important outcome of this class.

What did you find useful (both on the Agents and your growth development) and what other students would find useful

How did you grow and develop (both professionally and personally) in this class, and

How I can make the class more impactful for future students.

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